DEEP LEARNING

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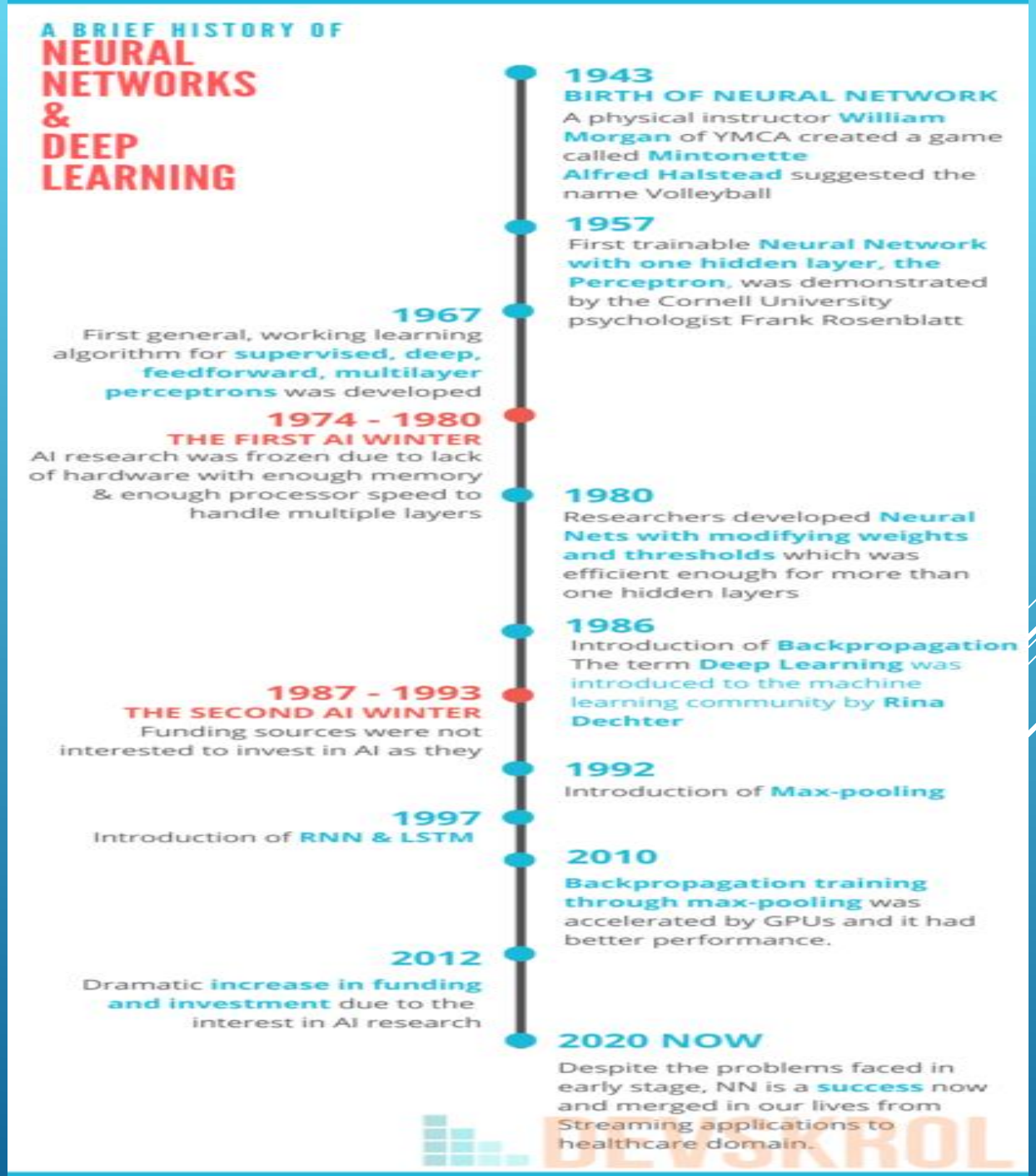
What is Deep Learning?

Deep Learning is a branch of **Machine Learning (ML)** that uses **Artificial Neural Networks (ANNs)** with many layers (hence "deep") to automatically learn patterns from data.
Unlike traditional ML algorithms (which often need manual feature extraction), **deep learning models learn features directly from raw data** such as images, text, or audio.

A brief history of Deep Learning:



How Deep Learning Works

- It uses **layers of artificial neurons**.
- Each layer transforms the input data and passes it to the next.
- The network automatically learns which features are important for prediction or classification.

Example:
In an image classification model:
- 1st layer learns edges
- 2nd layer learns shapes
- 3rd layer learns objects (like eyes, faces, etc.).

Deep Learning, is a more evolved branch of machine learning, and uses layers of algorithms to process data, and imitate the thinking process.

It is often used to visually recognize objects and understand human speech. Information is passed through each layer, with the output of the previous layer providing input for the next layer. The first layer in a network is called the input layer, while the last is called an output layer.

All the layers between input and output are referred to as hidden layers. Each layer is typically a simple, uniform algorithm containing one kind of activation function.

Applications of Deep Learning

- Image recognition (e.g., facial recognition, medical imaging)
- Natural language processing (e.g., ChatGPT, Google Translate)
- Speech recognition (e.g., Siri, Alexa)
- Autonomous vehicles
- Fraud detection, recommendation systems, etc.

Important Types of Deep Learning

1. Artificial Neural Network (ANN)

What it is:

- The **basic form** of a neural network.
- Inspired by how human brain neurons work.
- Data moves **in one direction** (input → hidden → output).

Structure:

- **Input layer:** Takes features (like study hours, age, etc.)
- **Hidden layers:** Process inputs using weights, bias, and activation functions.
- **Output layer:** Gives final result (e.g., pass/fail, yes/no).

Use Case Examples:

- Predicting sales, credit risk, or disease probability.
- Works with **structured/tabular data**.
- **Limitation:**
- Doesn't handle **images or sequences** efficiently.
- Every input feature is treated independently.

2. Convolutional Neural Network (CNN)

What it is:

- Designed specially for **image and video data**.
- Automatically extracts **spatial features** (edges → shapes → objects).

3. Recurrent Neural Network (RNN)

What it is:

- Designed for **sequential/time-dependent data**.
- Keeps **memory of previous inputs** — ideal for sentences, time series, etc.

Key Terms in Deep Learning

1. Neuron (Node)

•Basic unit of a neural network — similar to a brain cell.
•It takes inputs, applies weights, adds bias, and passes output through an activation function.

2. Layer

Neurons are organized in layers:
•**Input Layer:** Receives data features.
•**Hidden Layers:** Process data, learn features.
•**Output Layer:** Gives final result.

3. Weight

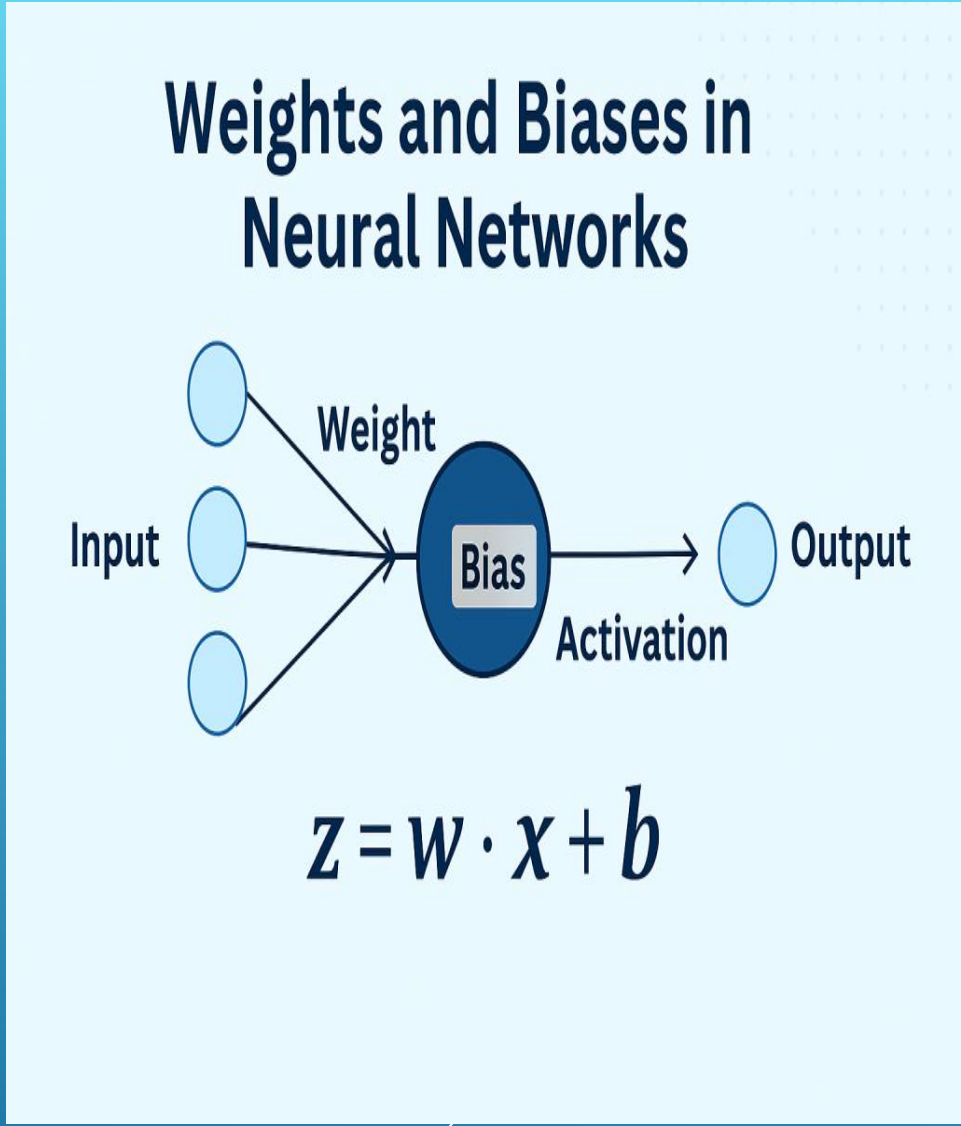
- A number that shows the **importance** of each input feature.
- During training, weights are adjusted to minimize error.

4. Bias

- Constant value added to shift activation function.
- Allows the neuron to fit data even when inputs are zero.

5. Activation Function

- Decides whether a neuron should activate or not.
- Introduces **non-linearity** so the network can learn complex patterns.



Z = w1 * x1 + w2 * x2 + b